Transfer: momentum and impulse

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Apply momentum reasoning to new collisions and deflection cases.

SUCCESS CRITERIA - I can...

1. Define the system and positive direction.
2. Use conservation or impulse only when the facts needed for the equation are true.
3. Explain the energy changes and the limits of your calculation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “impulse” to explain your answer. Sentence starters are optional.

momentum: mass times velocity, including direction

impulse: net force acting over time; change in momentum

conserved: unchanged in total under the stated conditions

01 Warm-up: bring back an earlier idea

In the same calculation for straight-line deflection, keep sideways Δv fixed but make lead time **3** times as long. **Compare shift** and **name a limitation**.

Word help: deflection = a change in the direction or path of motion

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

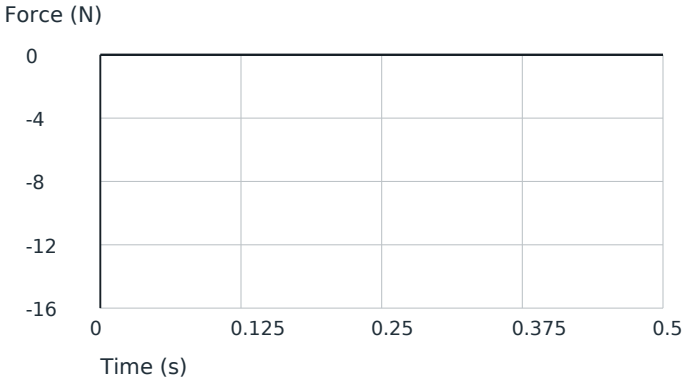
02 Worked example: explain the prediction

Reference: $p = mv$; $J = \Delta p = F_{avg}\,\Delta t$; $p_{total,\,i} = p_{total,\,f}$.
Use momentum conservation only when external impulse is negligible. **Label before**/after diagrams and directions.

Worked reasoning: A **2 kg** cart moving at **3 m/s** stops in **0.5 s**. Momentum change is **-6 kg·m/s**; average force is **-12 N**. Force direction opposes the initial motion.

Your turn: Represent the same stop with momentum arrows and a force-time rectangle.

Sentence starter: In my work, I used … to show … . I labeled it … .



Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

A **1 kg** cart at **4 m/s** right sticks to a **1 kg** cart at rest. **Draw before**/after and **explain which quantity** is conserved.

Word help: at rest = not moving in the chosen reference frame

Sentence starter: I think … . I can point to …, which shows …, so … .

Before contact: add velocity arrows.



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A **2 kg** cart moving right at **6 m/s** stops in **5 s**.
1. **Calculate its momentum-change magnitude** and average stopping-force magnitude.
2. Use the before-and-after diagram to **show the direction of the** change.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Before	After

Label the initial and final momentum.

Marzano check - circle: 1 2 3

05 Explain your reasoning

A report says a collision is elastic because total momentum is equal before and after. What extra calculation is needed to **check this conclusion**?

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** cart travels left at **6 m/s**. **Find momentum** and **draw its direction**.
Use before and after snapshots to organize the momentum account.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Before	After

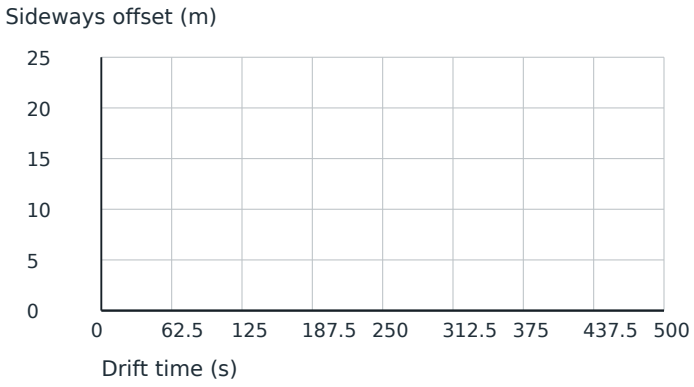
Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

A fixed impulse gives a target **0.04 m/s** sideways velocity change. **Find shift after 500 s** and **state a part of the** explanation that may not fit **what happens**.

Sentence starter: I think ... . I can point to ..., which shows ..., so ... .

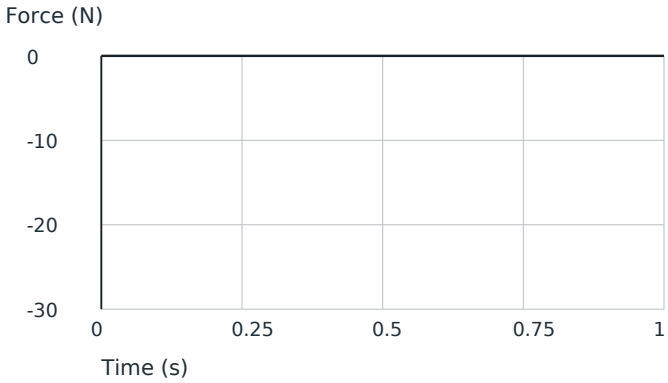


Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Return to the worked example in problem **2**.
1. Design two stopping processes for this cart that give the same impulse but different average forces.
2. **Explain the time-force tradeoff**.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3